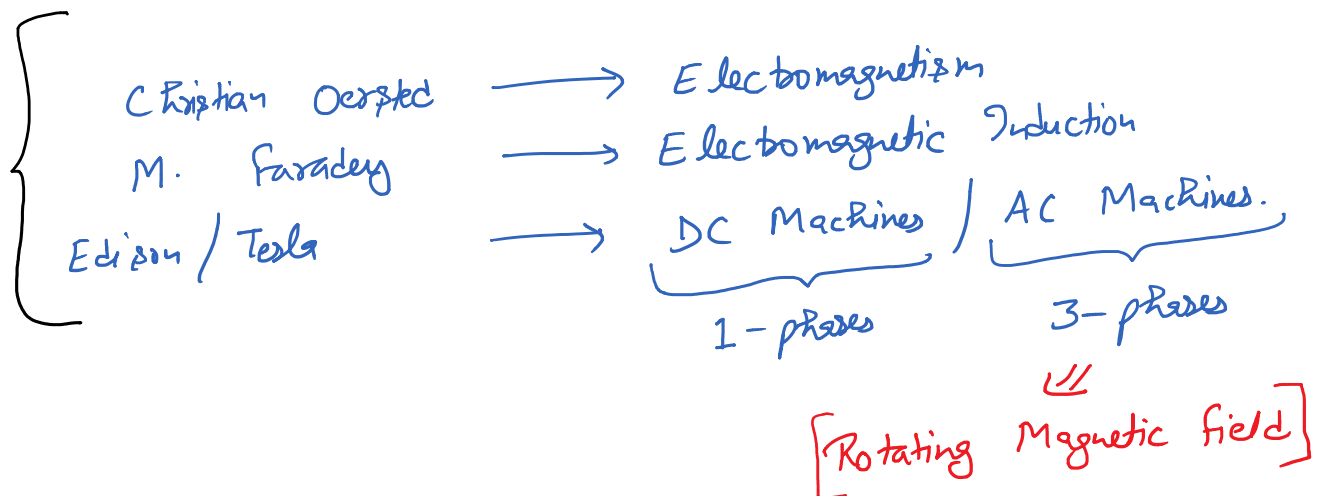
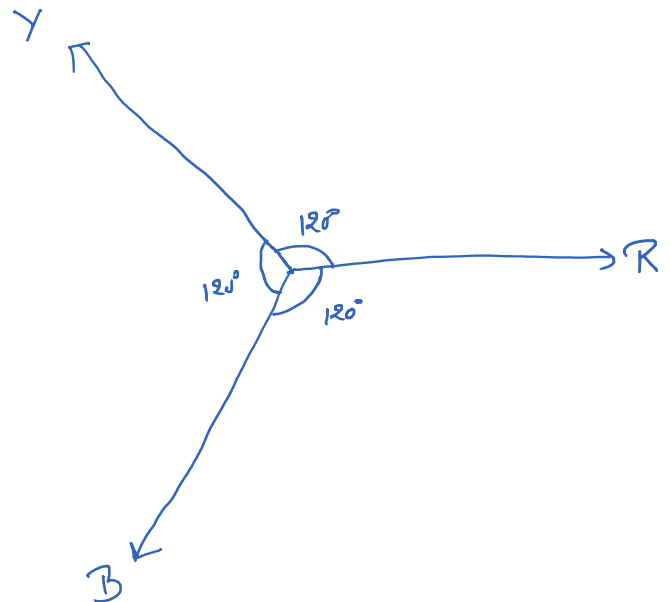


3 PHASE AC CIRCUITS

3 phases $\rightarrow$ R
 $\rightarrow$ Y
 $\rightarrow$ B



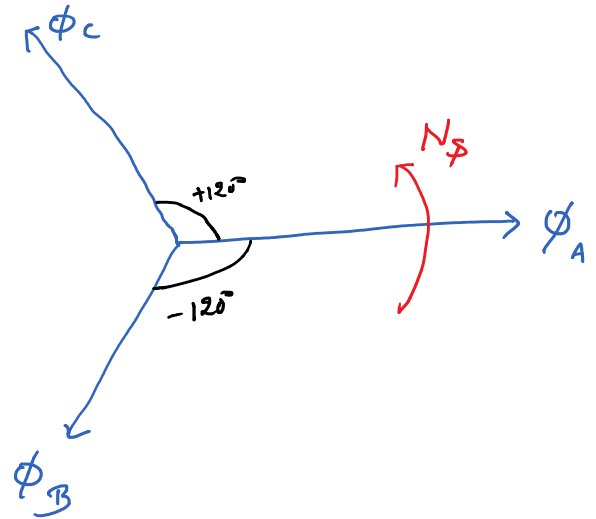
$\Rightarrow$ Rotating Magnetic field :-

When a 3 ϕ supply is given to 3 ϕ winding, it produces 3 ϕ current & 3 ϕ flux which finally produces one rotating magnetic field which runs at some constant speed (N_s) at some constant magnitude ($1.5 \phi_m$) at particular direction depending on the phase sequence.

3 ϕ supply	$\rightarrow$	120° time displacement
3 ϕ winding	$\rightarrow$	" space "

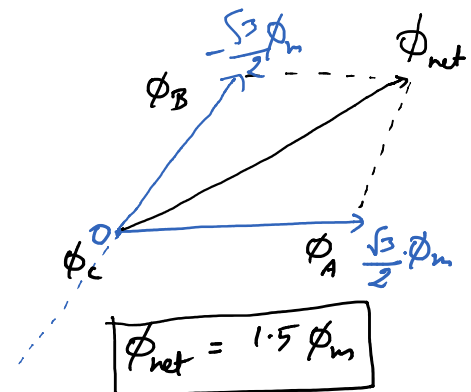
3 phase

$$\begin{aligned}\phi_A &= \phi_m \sin \omega t \\ \phi_B &= \phi_m \sin(\omega t - 120^\circ) \\ \phi_C &= \phi_m \sin(\omega t + 120^\circ)\end{aligned}$$



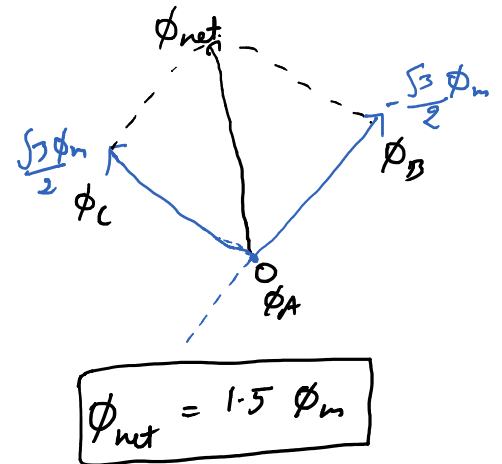
a) $\omega t = 60^\circ$

$$\begin{aligned}\phi_A &= \frac{\sqrt{3}}{2} \phi_m \\ \phi_B &= -\frac{\sqrt{3}}{2} \phi_m, \phi_C = 0\end{aligned}$$

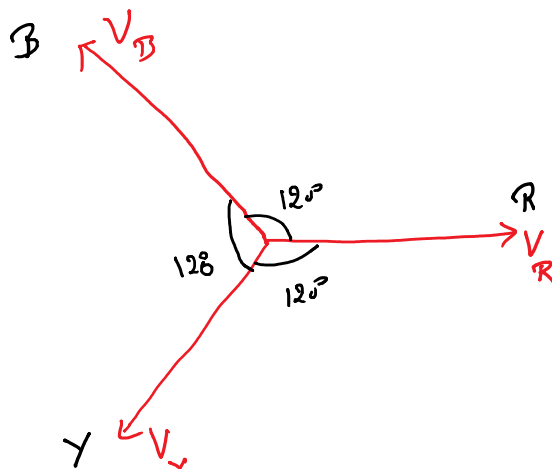


b) $\omega t = 0^\circ$

$$\begin{aligned}\phi_A &= 0, \phi_B = -\frac{\sqrt{3}}{2} \phi_m \\ \phi_C &= \frac{\sqrt{3}}{2} \phi_m\end{aligned}$$



$\Rightarrow$

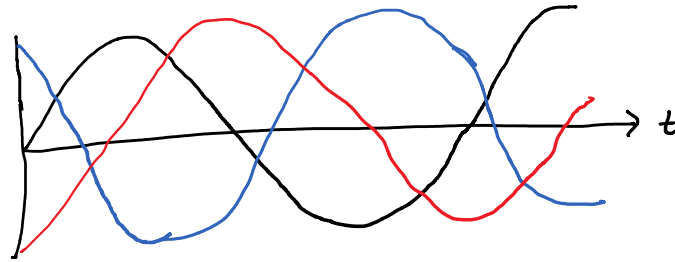


$$\begin{cases} \bar{V}_R = V_m \sin \omega t \\ \bar{V}_Y = V_m \sin(\omega t - 120^\circ) \\ \bar{V}_B = V_m \sin(\omega t - 240^\circ) \end{cases}$$

$\gamma \swarrow \searrow V_{\gamma}$

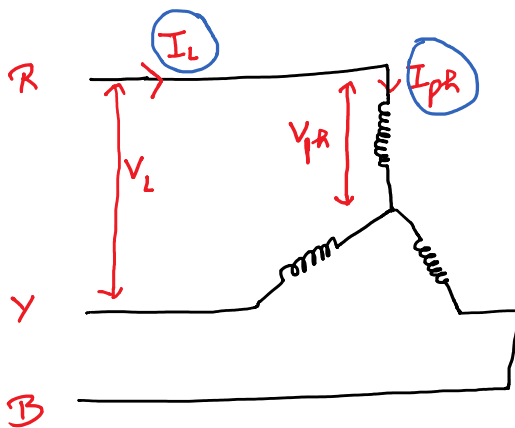
$$\bar{V}_B = V_m \sin(\omega t - 240^\circ)$$

$\swarrow$
+120°

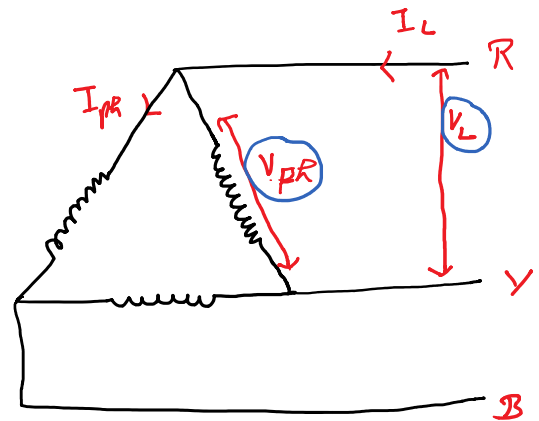


$\Rightarrow$

STAR (Y)



DELTA (Δ)



Balanced 3φ system / Concept of phase sequence

Any polyphase(n) system is said to be balanced :-

- (i) The magnitudes of corresponding quantities are equal in each phase
- &
- (ii) The phase difference between the corresponding quantities is given by:-

$$\theta = \frac{360^\circ}{n}$$

$$\boxed{\begin{matrix} n = 3 \\ \theta = 120^\circ \end{matrix}}$$

(only $n > 2$ i.e. $n \geq 3$)

ex:- $n = 3$ (3φ system)

(i) Magnitude Same

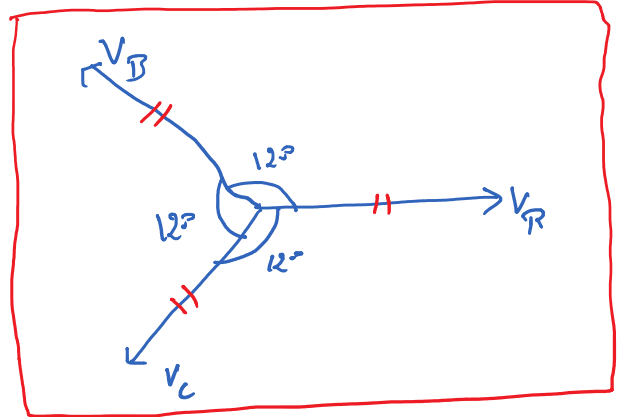
$$|V_a| = |V_b| = |V_c|$$

$$\text{---||---} \rightarrow V_a = V_R$$

$$\text{---||---} \rightarrow V_b = V_Y$$

$$\text{---||---} \rightarrow V_c = V_B$$

$$(ii) \theta = \frac{360^\circ}{3} = 120^\circ$$



3 ϕ balanced system.

Q $\Rightarrow$ The current in two phase system is given:-

$$i_a = \sqrt{2} \cdot I \cos(\omega t - \phi_1)$$

$$i_b = \sqrt{2} \cdot I \sin(\omega t - \phi_2)$$

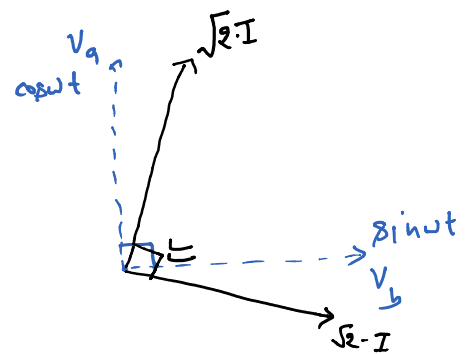
Find relationship b/w ϕ_1 & ϕ_2 so that i_a & i_b represents a two phase system.

Solⁿ:-

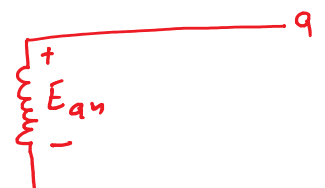
$$\left[\theta = \frac{360^\circ}{2} = 180^\circ \right]$$

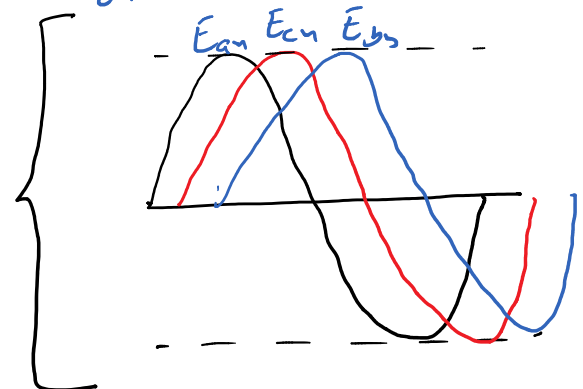
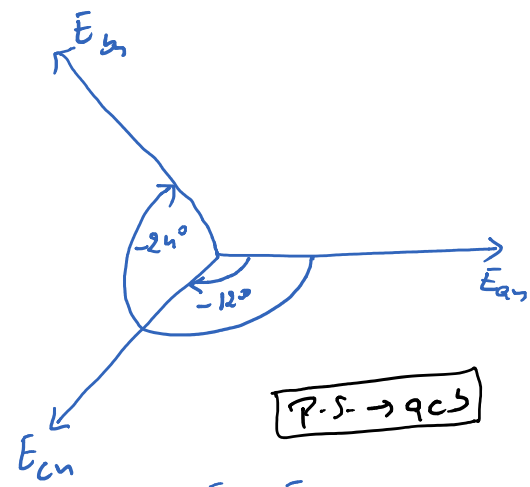
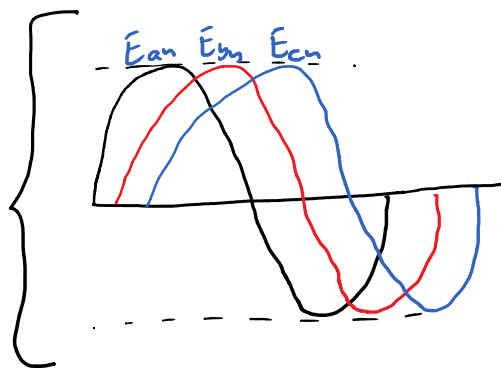
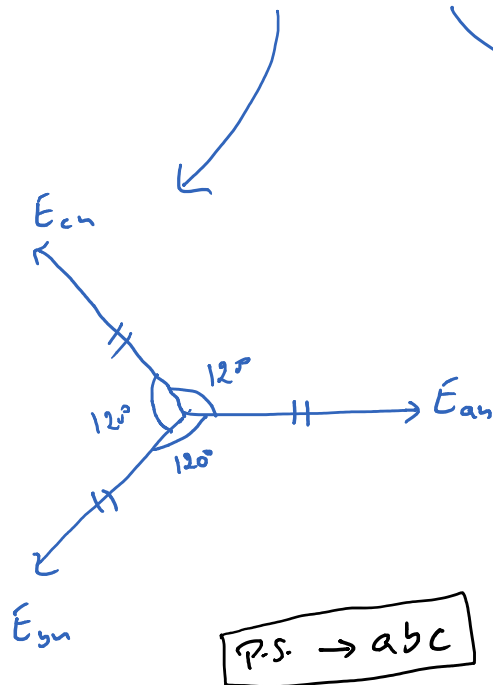
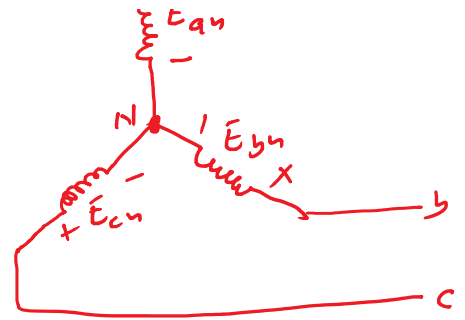
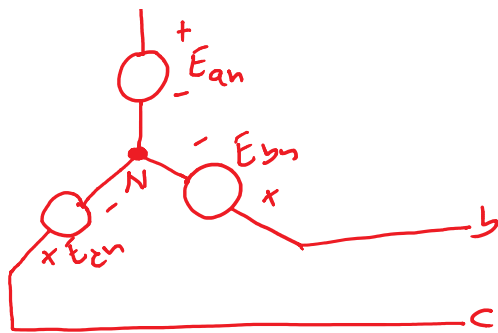
$$\left[\theta = 90^\circ \right]$$

$$|i_a| = |i_b|$$



Consider a balanced 3 ϕ ideal voltage source:-





$\Rightarrow$ phase sequence is defined as the order in which the phases attain their max. values.

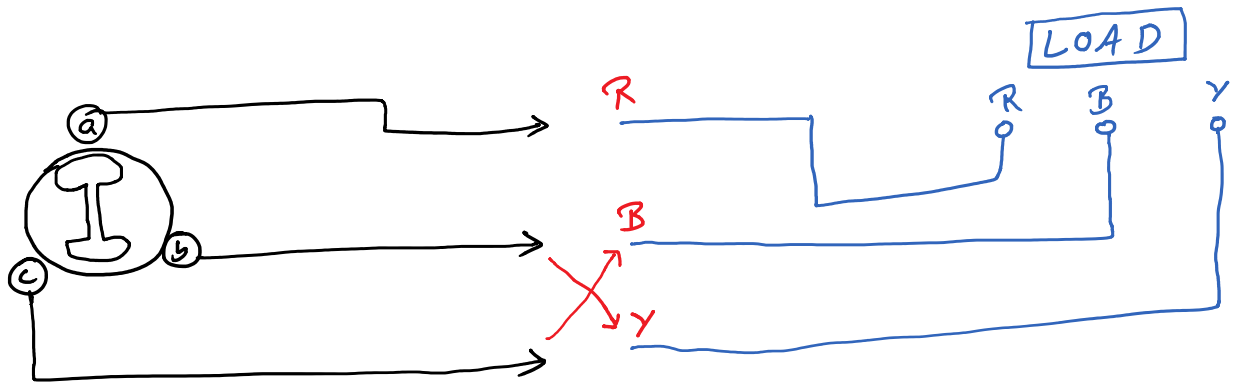
$\Rightarrow$ only two phase sequence is possible in 3 ϕ system.
(abc & acb)

$\Rightarrow$ Theoretically, P.S. can be reversed by reversing the rotation of rotor but practically not possible.

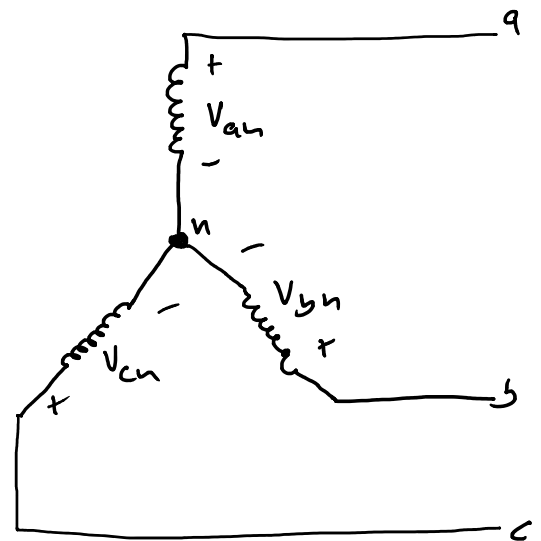
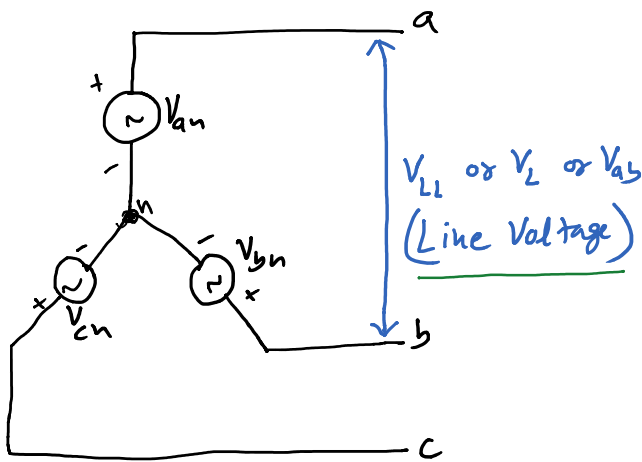
but,

... .. practically we can

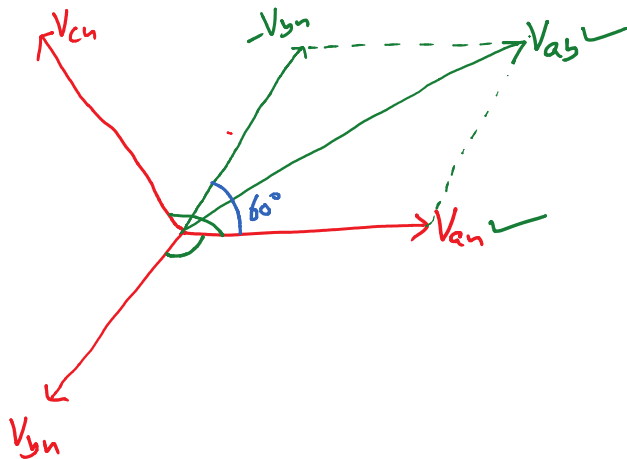
interchanging two terminals of a machine, power change the P.S. :-



Analysis of 3 ϕ Balanced Circuit :-



Balanced $\Rightarrow$ [3 ϕ AC Source] : $|V_{an}| = |V_{bn}| = |V_{cn}| = |V_{ph}|$



$$V_{an} = |V_{ph}| \angle 0^\circ$$

$$V_{bn} = |V_{ph}| \angle -120^\circ$$

$$V_{cn} = |V_{ph}| \angle +120^\circ$$

$$|V_{L3}|^2 = |V_{an}|^2 + |V_{bn}|^2 + 2|V_{an}||V_{bn}|\cos\theta$$

$$|V_{LL}|^2 = |V_{pR}|^2 + |V_{pR}|^2 + \cancel{2|V_{pR}||V_{pR}| \frac{1}{2}}$$

$$|V_{LL}|^2 = 3|V_{pR}|^2$$

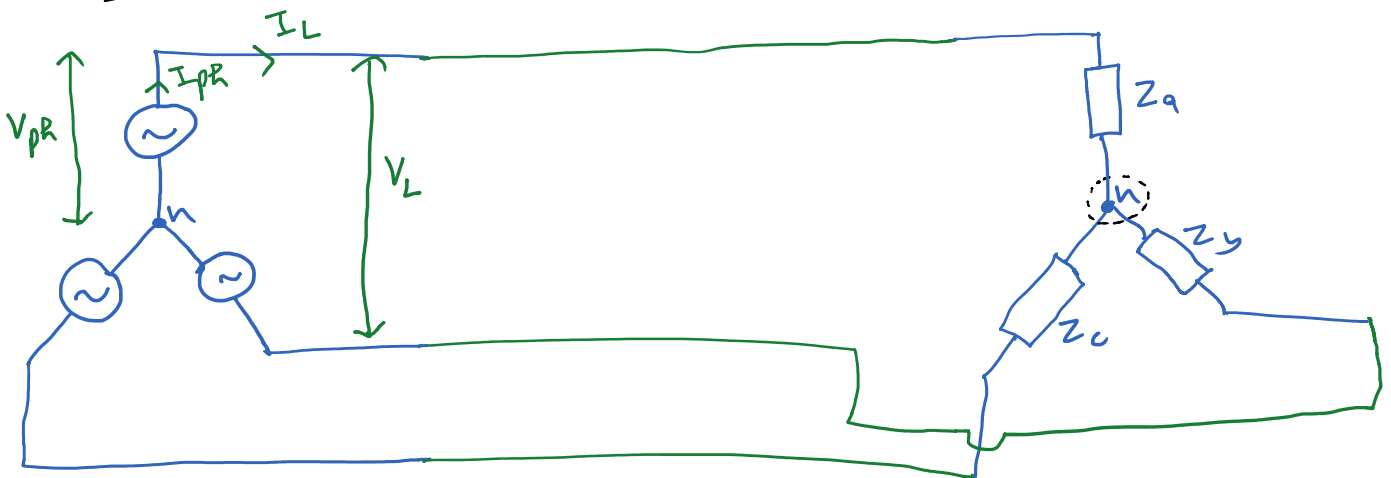
$$|V_{LL}| = |V_{ab}| = \sqrt{3} |V_{pR}|$$

Line Voltage

Phase voltage.

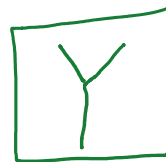
⇒ Line voltages are $\sqrt{3}$ times more in magnitude than the phase voltages. (In STAR connection system only)

⇒ 3 ϕ Y connected supply across 3 ϕ STAR connected Load



$$I_{pR} = I_L$$

$$V_L = \sqrt{3} \cdot V_{pR}$$



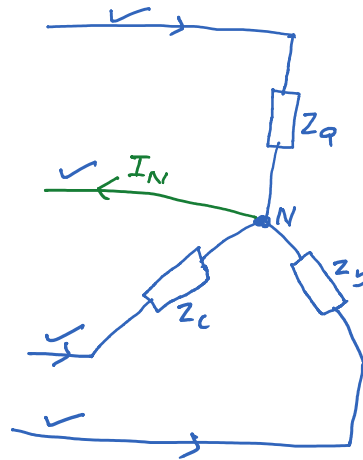
⇒ $|Z_1| = |Z_2| = |Z_3|$ } Load is balanced.

N → will be at zero potential.

⇒ 3 ϕ load may or may not be balanced.

then, we will be dealing with 3 ϕ 4 wire system., for proper

function of rotating machine, balancing is required.



$$\boxed{I_n \neq 0}$$

$$\Rightarrow I_a + I_b + I_c = I_N = 0 \quad \left. \vphantom{I_a + I_b + I_c = I_N = 0} \right\} \text{balanced } 3\phi$$

$$\begin{aligned} \Rightarrow V_a + V_b + V_c &= V_{ph} \angle 0^\circ + V_{ph} \angle 120^\circ + V_{ph} \angle -120^\circ \\ &= \underbrace{V_{ph}} + \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}j \right) \underbrace{V_{ph}} + \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}j \right) \underbrace{V_{ph}} \\ &= 0 \end{aligned}$$

for a 3 ϕ balanced system $\Rightarrow \begin{cases} \Sigma \bar{I} = 0 \\ \Sigma \bar{V} = 0 \end{cases}$

STAR CIRCUITS :-

$\Rightarrow$ if 3 ϕ supply is connected to 3 ϕ connected balanced load, 3 ϕ problem can be solved using 1 ϕ equations.





$$Z_a = Z_b = Z_c = Z$$

apply 1 ϕ concept :-

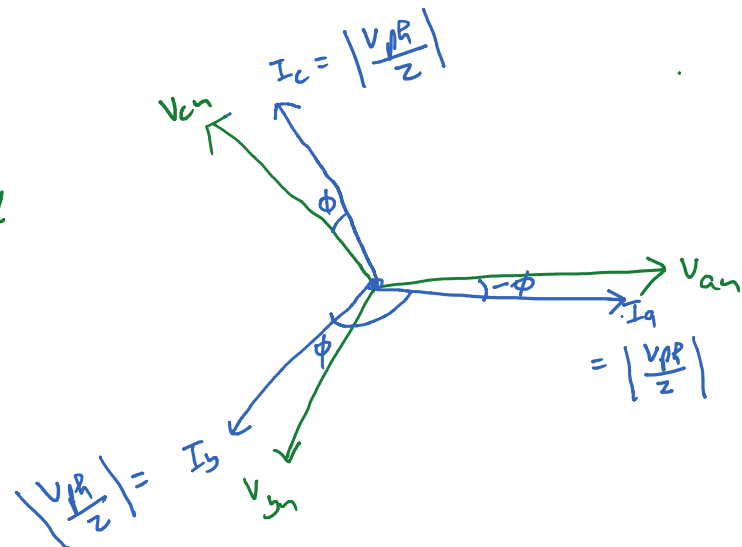
$$I_a = \frac{V_a}{Z} = \frac{V_{an}}{Z} = \frac{|V_{ph}| \angle 0^\circ}{|Z| \angle \phi}$$

$$\Rightarrow \checkmark I_a = \left| \frac{V_{ph}}{Z} \right| \angle -\phi$$

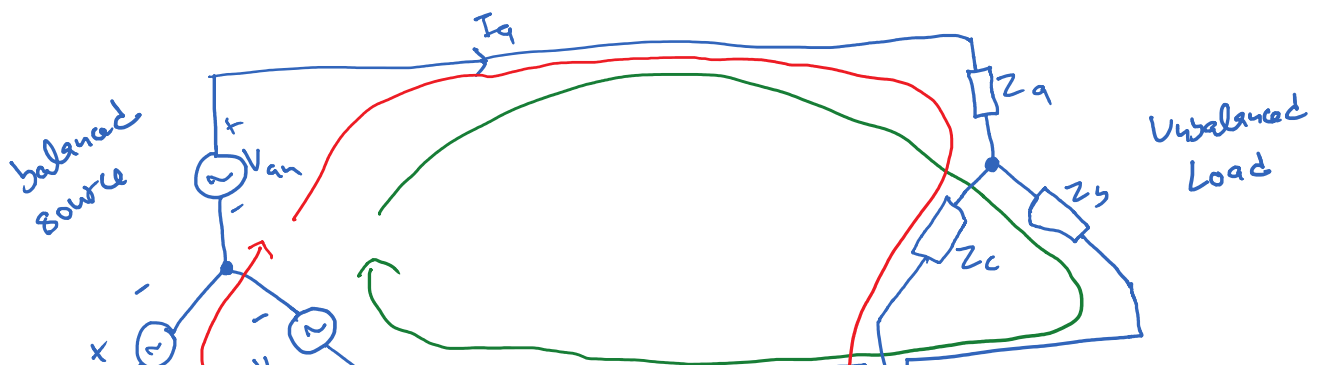
$$\Rightarrow \checkmark I_b = \frac{V_b}{Z} = \frac{V_{bn}}{Z} = \frac{|V_{ph}| \angle -120^\circ}{|Z| \angle \phi} = \left| \frac{V_{ph}}{Z} \right| \angle -120^\circ - \phi$$

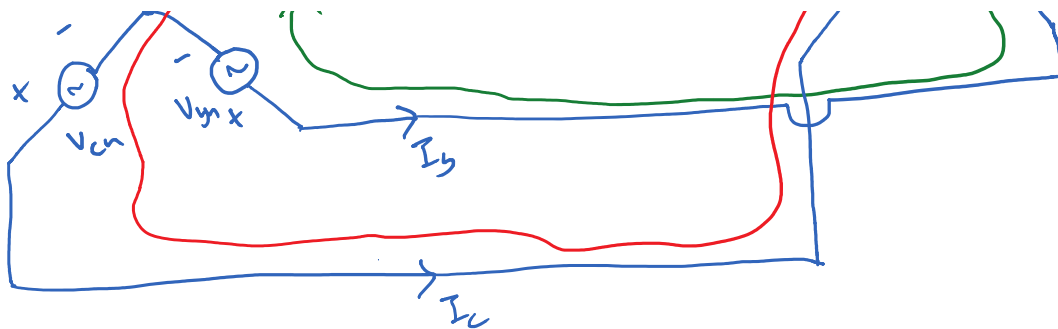
$$\Rightarrow \checkmark I_c = \frac{V_c}{Z} = \frac{V_{cn}}{Z} = \frac{|V_{ph}| \angle +120^\circ}{|Z| \angle \phi} = \left| \frac{V_{ph}}{Z} \right| \angle +120^\circ - \phi$$

$\Rightarrow$ balanced 3 ϕ supply is connected to Y balanced 3 ϕ line.



FOR UNBALANCED LOAD ($Z_a \neq Z_b \neq Z_c$ (or) $Z_a = Z_b \neq Z_c$)





$$-V_a + I_a Z_a - I_b Z_b + V_b = 0 \quad \text{--- (1)}$$

$$-V_a + I_a Z_a - I_c Z_c + V_c = 0 \quad \text{--- (2)}$$

$$I_a + I_b + I_c = I_n = 0 \quad \text{--- (3)}$$

using Millman's theorem:-

$$V_n = \frac{\frac{V_a}{Z_a} + \frac{V_b}{Z_b} + \frac{V_c}{Z_c}}{\frac{1}{Z_a} + \frac{1}{Z_b} + \frac{1}{Z_c}}$$

Millman's Theorem:-

$$R = R_1 + R_2 + R_3 + \dots$$

$$V = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \dots$$

$$\Rightarrow I_a = \frac{V_a - V_n}{Z_a}, \quad I_b = \frac{V_b - V_n}{Z_b}, \quad I_c = \frac{V_c - V_n}{Z_c}$$

$$I_a + I_b + I_c = 0.$$

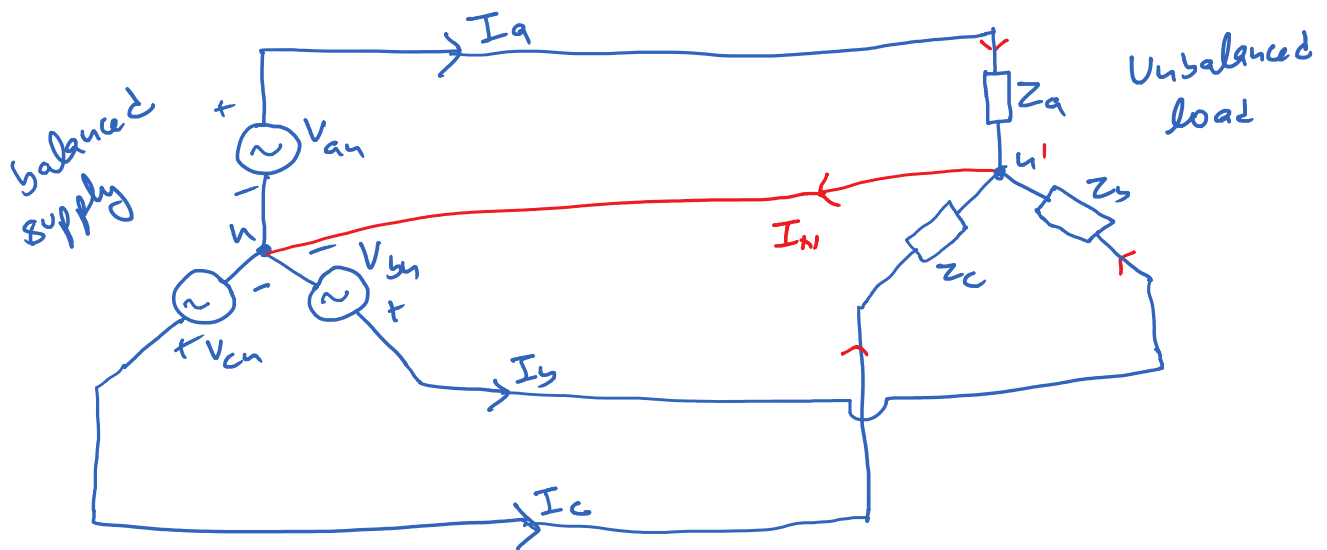
$\Rightarrow V_n \rightarrow$ indicate the degree of unbalancing.

Higher the value of V_n , larger the degree of unbalancing.

Neutral points connected :- (3 ϕ 4 wire system)



∴ balanced



$$I_a = \frac{V_a \angle 0^\circ}{Z_a}, \quad I_b = \frac{V_b \angle -120^\circ}{Z_b}, \quad I_c = \frac{V_c \angle +120^\circ}{Z_c}$$

$$I_a + I_b + I_c = I_N \neq 0$$

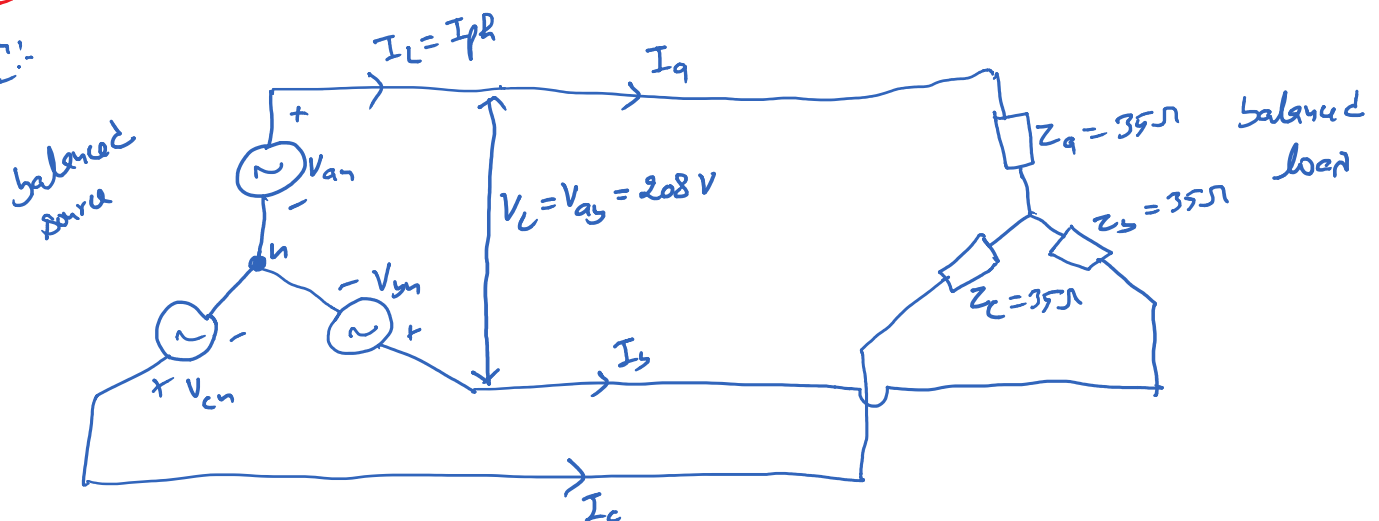
Q. A balanced 3 ϕ star connected load is fed from a 208 volt 3 ϕ supply. Each leg of the load has a resistance of 35 Ω .

find:-

- (i) Power factor
- (iii) Phase Currents

- (ii) Power consumed
- (iv) Line Currents.

Sol:-



$$\Rightarrow V_L = \sqrt{3} \cdot V_{ph} \quad \Rightarrow V_{ph} = \frac{208}{\sqrt{3}} = 120V.$$

$$|V_{an}| = |V_{bn}| = |V_{cn}| = |V_{ph}| = 120V.$$

$$\text{Phase voltages} \begin{cases} V_{an} = 120V \\ V_{bn} = 120 \angle -120^\circ V \\ V_{cn} = 120 \angle +120^\circ V \end{cases}$$

$$\text{Line Phase currents} \begin{cases} I_a = \frac{V_{an}}{R} = \frac{120 \angle 0^\circ}{35} = 3.428 \angle 0^\circ \\ I_b = \frac{V_{bn}}{R} = \frac{120 \angle -120^\circ}{35} = 3.428 \angle -120^\circ \\ I_c = 3.428 \angle +120^\circ \end{cases}$$

$$\Rightarrow P_{1\phi} = I^2 \cdot R = (3.428)^2 \times 35 \Rightarrow P_{1\phi} = 411.3 \text{ watts.}$$

$$\Rightarrow \boxed{P_{3\phi} = 3 \cdot P_{1\phi} \cong 1234 \text{ watts}}$$

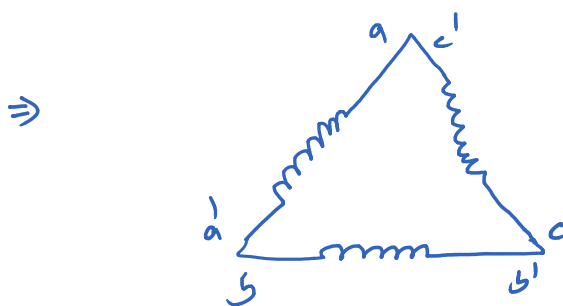
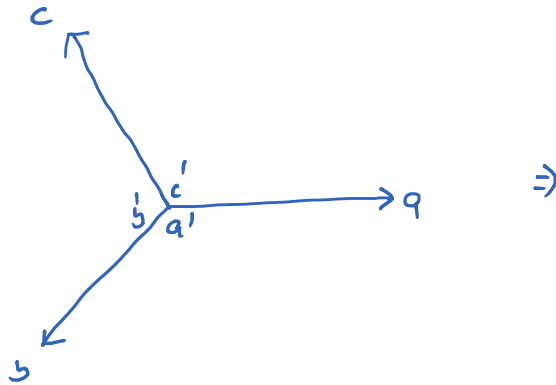
$$\Rightarrow \boxed{\text{Power factor} = 1 \text{ (unity)} \Rightarrow \text{Resistive n/w}}$$

$$\Rightarrow \boxed{|I_L| = |I_{ph}| = 3.428 \text{ Amp}}$$

Q. 43 Assume impedances are equal and having a value of $(15 + j20) \Omega$.

$$\begin{aligned} P_{3\phi} &\Rightarrow 1037 \text{ watt} \\ |I_L| &= |I_{ph}| = 4.8 \text{ Amp} \\ pf &= \cos \phi = \cos(53^\circ) \end{aligned}$$

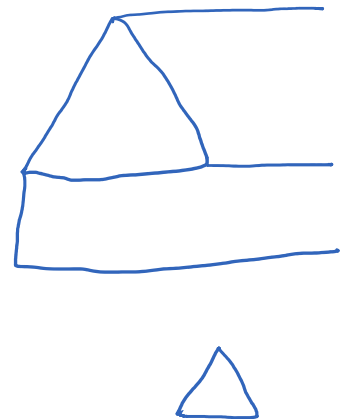
DELTA CIRCUITS :-



Most of the times, 3 ϕ supply is star connected. If mentioned specifically we consider delta connection.



[Insulation requirement increases $\Rightarrow$ $\uparrow$ cost]



$$V_{PR} = \frac{V_L}{\sqrt{3}}$$

$$I_{PR} = I_L$$

$\downarrow$
 $x_1 = 1A$

$\downarrow$
 $1A$

$$V_{PR} = V_L$$

$$I_{PR} = \frac{I_L}{\sqrt{3}}$$

$\downarrow$
 $\left(\frac{1}{\sqrt{3}}\right)$

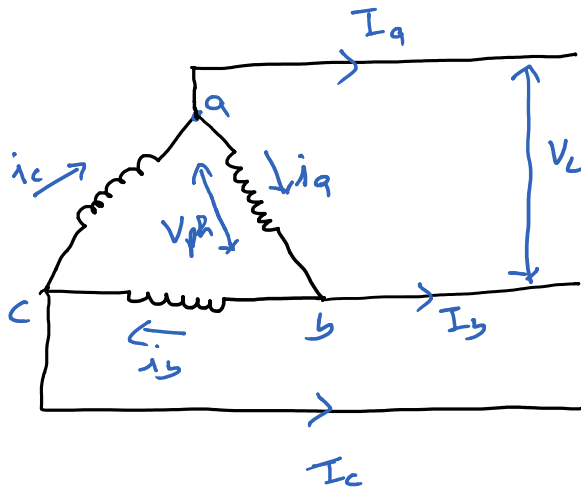
$\downarrow$
 $1A$

$I_A \angle$

$X_L = 1A$

I_A

$\left(\frac{1}{\sqrt{3}}\right)$

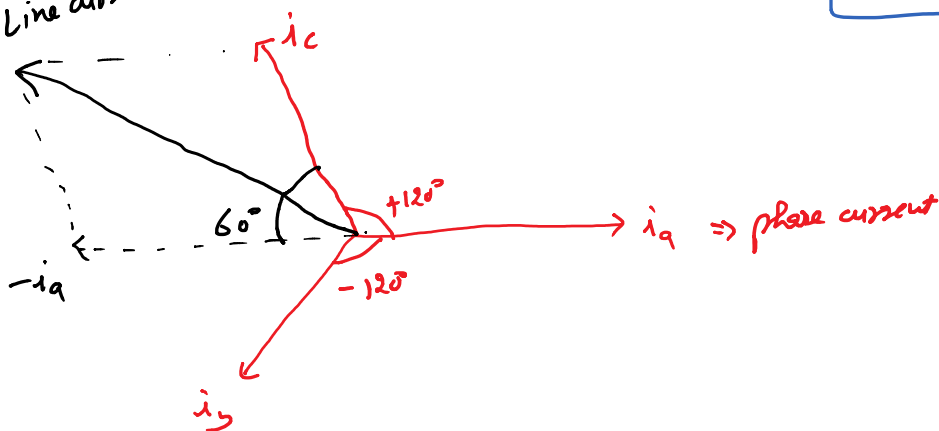


$I_a, I_b, I_c \rightarrow$ Line currents
 $i_a, i_b, i_c \rightarrow$ phase "

$$V_L = V_{PR}$$

$$I_a = i_c - i_b$$

$I_a \Rightarrow$ Line current



$$|I_a|^2 = |i_a|^2 + |i_c|^2 + 2|i_a||i_c|\cos\phi$$

$$|I_L|^2 = |I_{PR}|^2 + |I_{PR}|^2 + 2|I_{PR}| \cdot |I_{PR}| \cdot \frac{1}{2}$$

$$|I_L|^2 = 3|I_{PR}|^2$$

$$I_L = \sqrt{3} |I_{PR}|$$

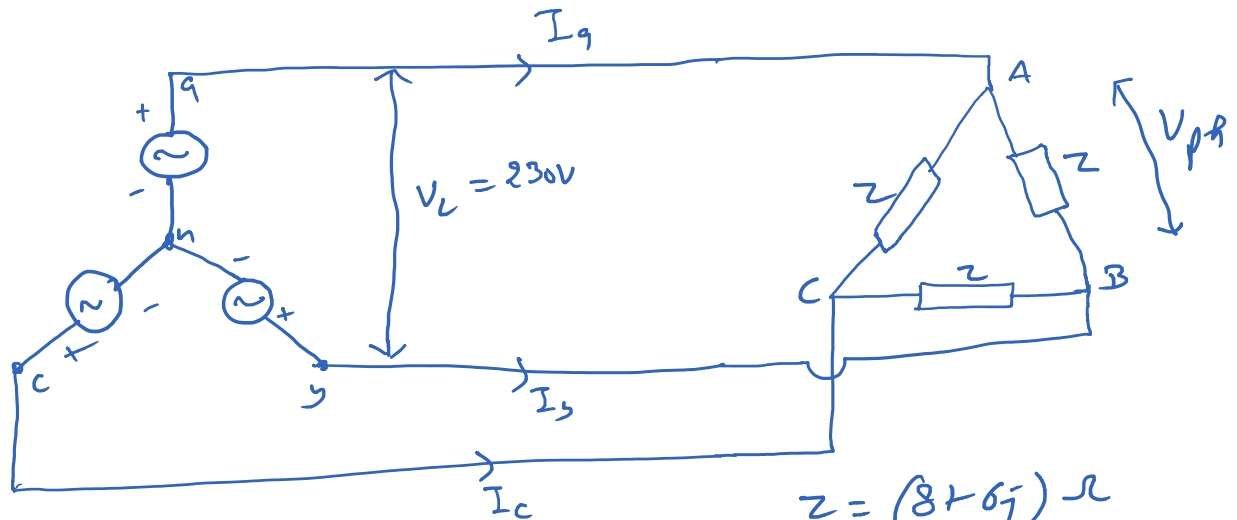
$\Downarrow$
Delta Circuit

- $\Rightarrow$ In Delta Circuits, line current is $\sqrt{3}$ times more than the phase current.
- $\Rightarrow$ In Delta circuits, line voltage is equal to phase voltage.

$\Rightarrow$ In Delta circuits, line voltage is equal to V_{ph}
 $\Rightarrow$ A balanced delta connected load of $(8+j6)\Omega$ per phase is connected to a 3 phase 230V supply. Find:-

- (i) Line current
 (ii) Power factor
 (iii) Power, Reactive Power & total volt-ampere

Sol:-



$$Z = (8 + j6)\Omega$$

$$|Z| = \sqrt{8^2 + 6^2} = 10$$

$$\phi = \tan^{-1}\left(\frac{6}{8}\right)$$

$$V_{ph} = V_L = 230V$$

$$\Rightarrow I_{ph} = i_a = \frac{V_{ph}}{Z} = \frac{230 \angle 0^\circ}{|Z| \angle \phi} = \frac{230}{10} \angle -\tan^{-1}\left(\frac{6}{8}\right)$$

$$\Rightarrow I_{ph} = 23 \angle -36.9^\circ \text{ Amp.}$$

$$|I_L| = \sqrt{3} (I_{ph}) = \sqrt{3} \cdot |i_a| = \sqrt{3} \cdot 23 = 39.8 \text{ Amp}$$

$$\therefore \text{Line currents} \begin{cases} I_a = 39.8 \angle 0^\circ \\ I_b = 39.8 \angle -120^\circ \\ I_c = 39.8 \angle +120^\circ \end{cases}$$

$$\text{currents } \begin{cases} I_L = 39.8 \angle +120^\circ \end{cases}$$

$$\therefore \text{power factor} = \cos \phi = 0.8$$

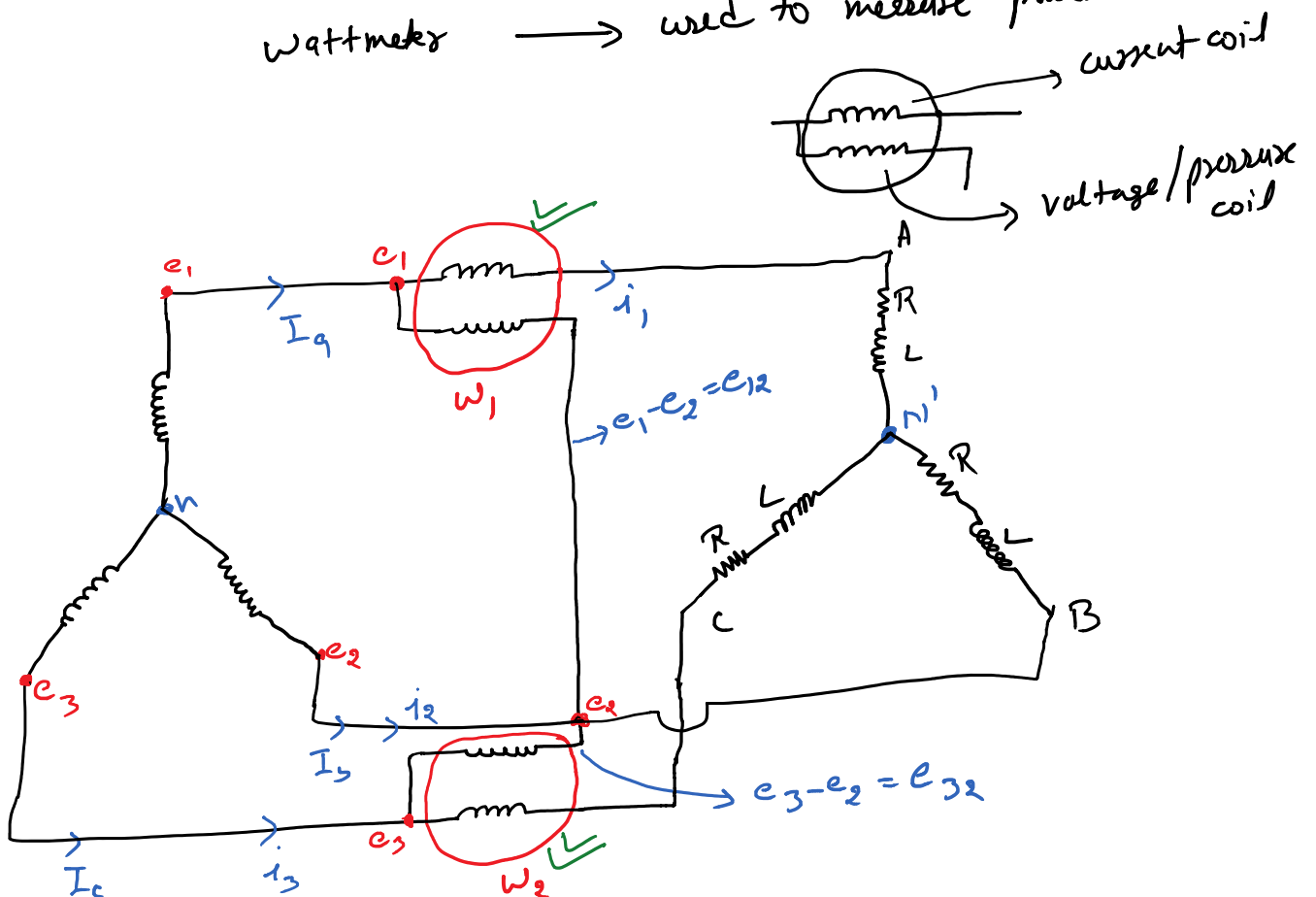
$$\begin{aligned} \text{Power} &= \sqrt{3} \cdot V_L \cdot I_L \cdot \cos \phi \\ &= \sqrt{3} \cdot (230) \cdot (39.8) \cdot (0.8) \approx 12.7 \text{ kWatt.} \end{aligned}$$

$$\begin{aligned} \text{Reactive Power} &= \sqrt{3} \cdot V_L \cdot I_L \cdot \sin \phi \\ &= \sqrt{3} \times 230 \times 39.8 \times \sin(\cos^{-1} 0.8) \approx 9.5 \text{ kV.} \end{aligned}$$

$$\begin{aligned} \text{Total V-A power} &= \sqrt{3} \cdot V_L \cdot I_L \\ &= \sqrt{3} \times 230 \times 39.8 = 15.8 \text{ kVA} \end{aligned}$$

Power Measurement (Two wattmeter method) :-

wattmeter $\rightarrow$ used to measure power



$$V_{pc}(w_1) = e_1 - e_2$$

$$V_{pc}(w_2) = e_3 - e_2$$

$$I_{cc}(w_1) = i_1$$

$$I_{cc}(w_2) = i_3$$

$$\Rightarrow (w_1 + w_2) = (e_1 - e_2)i_1 + (e_3 - e_2)i_3$$

$$= e_1 i_1 - e_2 i_1 + e_3 i_3 - e_2 i_3 \quad \text{--- (1)}$$

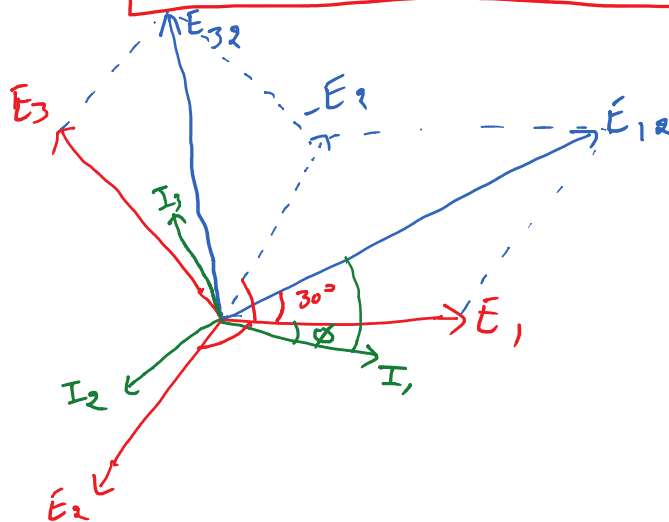
$$\Rightarrow i_1 + i_2 + i_3 = 0 \quad \Rightarrow i_2 = -(i_1 + i_3) \quad \text{--- (2)}$$

using eqⁿ (1) & (2) :-

$$w_1 + w_2 = e_1 i_1 + e_3 i_3 - e_2 (i_1 + i_3)$$

$$= e_1 i_1 + e_3 i_3 - e_2 (-i_2)$$

$$w_1 + w_2 = e_1 i_1 + e_2 i_2 + e_3 i_3$$



$$(p.f. = \cos \phi)$$

$$\bar{E}_{12} = \bar{E}_L = \bar{E}_1 - \bar{E}_2$$

$$|E_{12}| = \sqrt{3} |E_1|$$

$$E_L = \sqrt{3} \cdot (E_{pr})$$

$$W = w_1 + w_2 = E_{12} I_1 \cos(30 + \phi) + \bar{E}_{32} I_3 \cos(30 - \phi)$$

$$|E_{12}| = |E_{32}| = \sqrt{3} \cdot |E_{ph}|$$

$$\text{Let take } \Rightarrow |I_1| = |I_3| = (I)$$

$$\begin{aligned} W &= \sqrt{3} E_{ph} I \cos(30^\circ) + \sqrt{3} E_{ph} I \cos(30^\circ - \phi) \\ &= \sqrt{3} E_{ph} I [\cos(30^\circ + \phi) + \cos(30^\circ - \phi)] \\ &= \cancel{2} \cdot \sqrt{3} \cdot E_{ph} \cdot I \cdot \cancel{\cos 30^\circ} \cdot \cos \phi \\ &\quad \quad \quad \frac{\sqrt{3}}{2} \end{aligned}$$

$$\boxed{W = 3 E_{ph} \cdot I \cdot \cos \phi} \quad \star$$

$$[W_1 + W_2 = 3 E_{ph} I \cos \phi] \quad \text{--- (A)}$$

Similarly:-

$$W_2 - W_1 = \sqrt{3} E_{ph} I [\cos(30^\circ + \phi) - \cos(30^\circ - \phi)]$$

$$[W_2 - W_1 = \sqrt{3} E_{ph} I \sin \phi] \quad \text{--- (B)}$$

from (A) & (B):-

$$\frac{W_2 - W_1}{W_2 + W_1} = \frac{\sin \phi}{\sqrt{3} \cos \phi}$$

$$\Rightarrow \left[\tan \phi = \frac{\sqrt{3} (W_2 - W_1)}{W_1 + W_2} \right]$$

$$\phi = \cos^{-1} \left[\frac{\sqrt{3} (W_2 - W_1)}{W_1 + W_2} \right]$$

$$P.f. = \cos \phi = \cos \left[\tan^{-1} \left(\frac{\sqrt{3} (W_2 - W_1)}{W_2 + W_1} \right) \right]$$

Q $\Rightarrow$ (44) A 500 volt 3-phase motor has an output of 37.3 kW and operates at a power factor of 0.85 with an efficiency of 90%. Calculate the reading on each of the two wattmeter connected to measure the input.

$$\begin{aligned} W_1 &= 13.31 \text{ kW} \\ W_2 &= 28.13 \text{ kW} \end{aligned}$$

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